Analytical Cumulative Examination - 13th November 1993

TABLE 4. Elements determined by RNAA in Noblesville and other H chondrites.

Element	Noblesville	H4-6 chondrites	(No.)†
Au (ppb)	187	210 ± 48	(26)
Co (ppm)	567	790 ± 180	(58)
Ga (ppm)	6.20	5.9 ± 0.8	(55)
Rb (ppm)	3.10	2.2 + 1.1 − 0.7	(58)
Sb (ppb)	90	77 + 38 − 25	(45)
Ag (ppb)	38.9	34 + 37 − 18	(58)
Se (ppm)	8.71	8.1 ± 1.2	(58)
Cs (ppb)	131	32 + 105 − 25	(58)
Te (ppb)	401	360 ± 120	(58)
Zn (ppm)	53.9	48 ± 17	(58)
Cd (ppb)	1.9	4.8 + 29.8 − 4.1	(58)
Bi (ppb)	3.35	1.6 + 3.7 − 1.1	(58)
Tl (ppb)	1.90	0.33 + 1.73 − 0.05	(58)
In (ppb)	3.30	0.43 + 1.05 − 0.30	(58)

† Data from Lingner *et al.* (1987) and Wolf and Lipschutz (1992). Arithmetic means and associated standard deviations are indicated if one uncertainty value is listed. Otherwise, the mean is a geometric one.

Table 5. Radionuclides in Noblesville and other H chondrite falls.

Isotope	Half-life	Noblesville (dpm/kg)	H Chondrites (dpm/kg)	Chondritic Range (dpm/kg)
^{7}Be	(53.28 d)	61 ± 10		
^{10}Be‡	(1.5 Ma)	19.1 ± 1.3	20.6 ± 1.0§	
^{22}Na	(2.601 y)	131 ± 15	60 – 110	60 – 80
^{26}Al	(0.7 Ma)	52 ± 3	56.1 ± 1.0§	
^{26}Al‡	(0.7 Ma)	47.8 ± 1.4		
^{36}Cl‡	(0.301 Ma)	22.5 ± 1.3*	22.8 ± 3.1*	
^{46}Sc	(83.8 d)	8 ± 1	5 – 15	~8
^{48}V	(15.97 d)	20 ± 10	15 – 40	~16
^{51}Cr	(27.71 d)	27 ± 5	50 – 130	
^{54}Mn	(312.5 d)	99 ± 11	60 – 110	60 – 105
^{56}Co	(78.5 d)	8 ± 2	5 – 15	
^{57}Co	(271 d)	12 ± 2	5 – 30	
^{58}Co	(71.3 d)	2 ± 1	2 – 25	
^{60}Co	(5.272 y)	0.9 ± 0.7	0 to > 100	
^{40}K	(1.28 Ga)	680 ± 69		

† Near solar maximum data from Evans *et al.* (1982), Evans and Reeves (1987).
‡ Determined by AMS. All other nuclides quantified by γ-ray counting.
* Units are dpm/kg of metal (Vogt, 1990). All other data are for bulk rock.
§ Uncertainties are one population standard deviation.

The two tables above were copied from the analytical section of a paper published this year describing a new meteorite fall, Noblesville, Indiana.

1. Nuclear analytical methods were used to determine the data in each table.

(a) Briefly describe the RNAA (radiochemical neutron activation analysis) method referred to in Table 4. In particular, stress the major differences between this method and similar methods available. (30 points)

(b) Describe the method or methods used to determine the data in Table 5. Credit will be given for detail. (30 points)

2. Explain the following points of detail.

(a) What is meant by arimethic means, associated standard deviations, and geometric means. (10 points)

(b) The data in Table 5 are given in units of dpm/kg instead of concentrations. Why? (5 points) Calculate the concentration of Co-56 in units of ug/g (ppm). (10 points)

(c) Why did the authors' mention that certain data were obtained near solar maximum? (10 points)

(d) Why is Cl-36 reported in terms of kg of metal instead of bulk rock? (5 points)